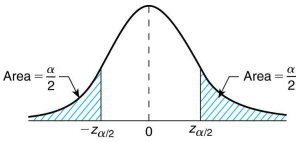
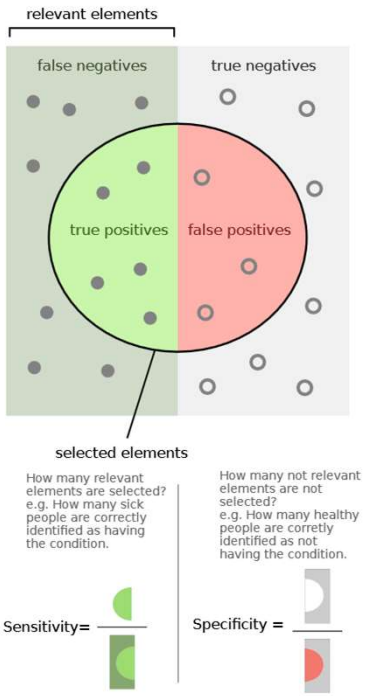
Agenda

- Earlier
  - Significance testing
- Today
  - Hypothesis testing
  - Type II errors
  - Qualitative variables and goodness-of-fit



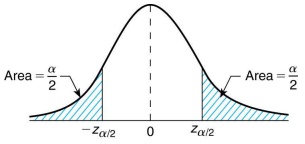


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Type I Errors

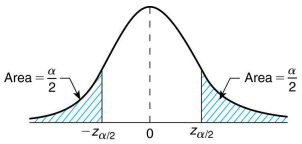
- If the null hypothesis is true, and we decide to reject
  - at the 90% confidence level, we will make an error 5% of the time
  - At the 95% level, we would be less wrong

Decision Making	
	Actual situation
Decision made	H_0 is true
Do not reject H0	OK
Reject H_0 as false	Type I error



3

Review



- 5 signals (possibly corrupted) are received by a transmitter on a particular day
  - mean value of $\bar{X} = 9.5$
- Null hypothesis: signal has mean 8, noise is $\mathcal{N}(0,4)$
- Test statistic $\text{se}(\bar{x}) = 2/\sqrt{5}, t_0 = (9.5 - 8)/\text{se}(\bar{x}) = 1.68$, two-sided p-value = $2\times\big(1-\text{normcdf}(1.68)\big) = 0.0936$
  - 90% conf ($\alpha/2 = 5$): 1.68 > 1.645, 0.095 < 0.10, CI = [8.03, 10.97], reject
  - 95% conf ($\alpha/2 = 0.025$): 1.68 < 1.96, 0.094 > 0.05, CI = [7.75, 11.25], do not reject
- If the null hypothesis was true, then we would be wrong in one case

$$t_o = \frac{\hat{\theta} - \theta_0}{\text{se}(\hat{\theta})}$$

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Example considering only one population

- We measuring sulfhydryl concentration (mmol) in blood samples
- Recall the numbers for 2-sided 95% confidence of the sample mean
  - n = 7, df = 6, t(0.025, 6) = 2.447 SE = 0.07559/√7 = 0.02857, margin = 2.447 × 0.02857 = 0.0699, CI = 1.92143 ± 0.0699 = (1.852, 1.991)
- A new patient has x_n = 2.10. Can we reject: patient comes from the without-RA population?
- The CI for the *mean* is the wrong measure

Without RA
1.84
1.92
1.94
1.92
1.85
1.91
2.07
Sample size 7
Sample mean 1.92143
Sample standard deviation 0.07559

5

Example

- Recall the numbers
  - n = 7, df = 6, $\bar{x}$ = 1.92143, t(0.025, 6) = 2.447, se = 0.07559/√7 = 0.02857, margin = 2.447 × 0.02857 = 0.0699,CI = 1.92143 ± 0.0699 = (1.852, 1.991)
- x_n = 2.10. H_0: new patient belongs to the without RA group
- We must account for
  - Variation of the new individual around the population mean
  - Uncertainty in our estimate of the mean
  - var$(x_n - \overline{x_1})$ = $0.07559^2 + 0.07559^2/7$, se = 0.0808, $t_o = \frac{2.10-\bar{x}}{0.0808}$ = 2.21, pvalue = 0.069
  - Inference: we fail to reject the null hypothesis

$$t_o = \frac{\hat{\theta} - \theta_0}{\mathrm{se}(\hat{\theta})}$$

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Example

- Recall the numbers
  - n = 7, df = 6, $\bar{x}$ = 1.92143, t(0.025, 6) = 2.447, se = 0.07559/√7 = 0.02857, margin = 2.447 × 0.02857 = 0.0699,CI = 1.92143 ± 0.0699 = (1.852, 1.991)
- x_n = 2.10. H_0: new patient belongs to the without RA group
- We must account for
  - var$(x_n - \overline{x_1})$ = $0.07559^2 + 0.07559^2/7$, se = 0.0808, $t_o = \frac{2.10-\bar{x}}{0.0808}$ = 2.21, pvalue = 0.069
  - Inference: we fail to reject the null hypothesis
- Equivalent form: the 95% prediction interval: 1.92143 ± 2.447 × 0.0808 = (1.724, 2.119). The value 2.10 is inside it.

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Significance Testing with Difference in Means

- $H_0: \mu_2 - \mu_1 = 0. H_1: \mu 2 - \mu 1 \neq 0$ α = 0.05
  - small values of n, do not have true variance
  - T-distribution, conservative df = min(n1−1, n2−1) = 5.

$$t_o = \frac{\hat{\theta} - \theta_0}{\mathrm{se}(\hat{\theta})}$$

- Difference between means numbers
  - Diff = 2.42 – 1.92143 = 0.49857, se = sqrt(0.07559²/7 + 0.44049²/6) = 0.1821
  - $t_0 = 0.49857/0.1821 = 2.74$
  - P-value = 0.041
- We reject the null hypothesis since 0.04 < 0.05
  - Note 2.74 exceeds 2.571 = t(0.025, 5) = 2.571
  - CI for diff: (0.0304, 0.9668) does not contain 0

	Without RA	With RA
	1.84	1.765
	1.92	3.015
	1.94	2.575
	1.92	2.225
	1.85	2.225
	1.91	2.715
	2.07	
Sample size	7	6
Sample mean	1.92143	2.42000
Sample standard deviation	0.07559	0.44049

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Review

- What is the essential difference between the information provided by a confidence interval and by a significance test?
- What is the most common form of null hypothesis?
- How does a null hypothesis typically relate to a belief, hunch, or research hypothesis that initiates a study?
- How can researchers try to demonstrate that effects or differences seen in their data are real?
- How does the alternative hypothesis typically relate to a belief, hunch, or research hypothesis that initiates a study?

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Review: One-side alternatives

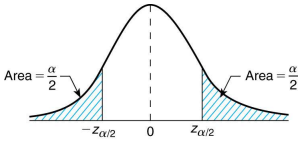
- We also use ***one-sided alternatives***
- Either: $H_1: \theta > \theta_0$ or $H_1: \theta < \theta_0$
  - when the research hypothesis specifies the direction of the effect
  - or more generally, when the investigators had good grounds for believing the true value of θ was on one particular side of θ_0 before the study began.
- Otherwise, a two-sided alternative $H_1: \theta \neq \theta_0$ is used.

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Type I Errors

- If the null hypothesis is true, and we decide to reject
  - at the 95% confidence level, we will make an error 5% of the time
  - At the 99% level, we would be less wrong
  - Never reject, we would never be wrong

Decision Making	
	Actual situation
Decision made	H_0 is true
Do not reject H0	OK
Reject H_0 as false	Type I error



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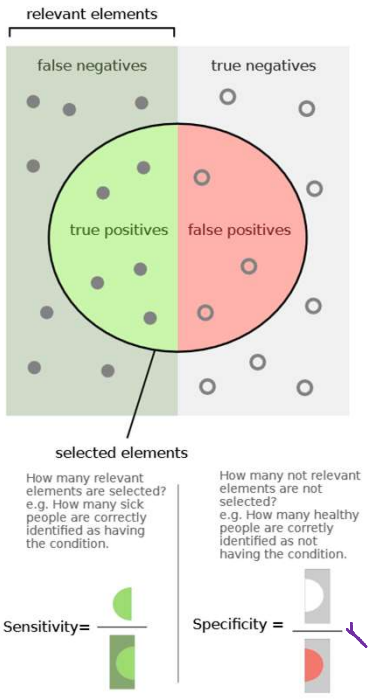
Type I and Type II

Decision Making		
	Actual situation	
Decision made	H_0 is true	H_0 is false
Do not reject H0	OK	Type II error
Reject H_0 as false	Type I error	OK

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Sensitivity and specificity

- Sensitivity (true positive ratio): probability of a positive test result, conditioned on the individual truly being positive
  - "recall" in machine learning
- Specificity (true negative ratio): probability of a negative test result conditioned on the individual truly being negative
- The tradeoff is explored in ROC analysis
- Confusion matrix depends on the field of research



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Type I and Type II errors

- There can be two types of errors
  - Type I: Incorrectly recommends rejection of H_0 (even if it is true)
  - Type II: Incorrectly fail to reject H_0 (even if it is false)
- Ideally, H_0 should be rejected only if the samples are very unlikely when H_0 is true
  - This calls for increasing sample size

The power of a test = 1 - probability of type II error

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Review: Type I and Type II errors

- There can be two types of errors
  - Type I: Incorrectly recommends rejection of H_0 (even if it is true)
  - Type II: Incorrectly fail to reject H_0 (even if it is false)
- Ideally, H_0 should be rejected only if the samples are very unlikely when H_0 is true.
- Review:
  - We design the test in such a way that whenever H_0 is true, the probability of rejection based on the samples (type I error) is never greater than α, called the significance level of the test.
  - It is a parameter set in advance. Typical values of α are 0.05 or 0.01.

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Type II error

- What is the probability of failing to reject the null hypothesis H_0 when the true mean μ is **unequal** to μ_0?
- Define $\beta(\mu)$ to be the probability of not rejecting H_0

$$\beta(\mu) = P_\mu\left\{\left|\frac{\overline{X}-\mu_0}{\sigma/\sqrt{n}}\right| \le z_{\alpha/2}\right\}$$
$$= P_\mu\left\{-z_{\alpha/2} \le \frac{\overline{X}-\mu_0}{\sigma/\sqrt{n}} \le z_{\alpha/2}\right\}$$

$$\beta(\mu) = \Phi\left(\frac{\mu_0-\mu}{\sigma/\sqrt{n}} + z_{\alpha/2}\right) - \Phi\left(\frac{\mu_0-\mu}{\sigma/\sqrt{n}} - z_{\alpha/2}\right)$$

The function $\beta(\mu)$ is called the **operating characteristic (OC) curve**. μ is hidden in the formula under the assumption that $\overline{X}$ is normal i.e., $Z = (\overline{X}-\mu)/(\sigma/\sqrt{n})$ is standard normal

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